

UNIVERSITY OF LONDON
INTERNATIONAL PROGRAMMES

BSc Computer Science and Related Subjects



CM3070 PROJECT
PRELIMINARY PROJECT REPORT

<Project Idea 4.2: Financial Advisor Bot (AI-Powered Stock
Recommendation System)>

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Date of Submission: 29/06/2026

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CHAPTER 1: INTRODUCTION

1.1 Project Concept

This project develops an AI-Powered Financial Advisor Bot for the stock market. The bot analyses financial data to make personalised recommendations for investment suitability, combining machine learning-driven stock signal prediction with a scenario-based investor risk profiling questionnaire. The system addresses a gap in retail investment tools: most stock screeners provide generic signals without considering the individual investor's risk tolerance, emotional comfort with volatility, or investment time horizon.

The system comprises four core components:

1. Risk Questionnaire — A 5-question scenario-based assessment (informed by Grable & Lytton, 1999) that classifies investors into one of five categories: Conservative, Moderately Conservative, Balanced, Growth, or Aggressive.
2. Stock Recommender — A Random Forest model trained on 5 years of historical data for 10 major stocks, predicting Buy/Hold/Avoid signals from 9 technical indicators (RSI, MACD, moving averages, volatility).
3. Suitability Engine — A deterministic mapping that combines the investor's risk category with each stock's signal to produce Suitable, Use Caution, or Not Suitable ratings with human-readable explanations.
4. Walk-Forward Backtesting — A temporal validation system that evaluates how the advisor's recommendations would have performed historically, providing evidence of recommendation quality.

1.2 Motivation

Retail investors increasingly make self-directed trading decisions using platforms like Robinhood, Trading 212, and eToro. Research shows that individual investors frequently make suboptimal decisions due to behavioral biases and a mismatch between their risk tolerance and chosen investments (Barber and Odean, 2013). Professional wealth managers provide personalised advice but their services are inaccessible to retail investors with smaller portfolios. This project democratises basic investment suitability assessment through an automated, transparent, AI-driven system.

1.3 Aims and Objectives

Aim: To develop and evaluate an AI financial advisor bot that provides personalised stock suitability recommendations by integrating scenario-based investor risk profiling with machine learning-based stock signal prediction and walk-forward backtesting evaluation.

#	Objective	Deliverable
O1	Design a scenario-based risk tolerance questionnaire with 5	src/risk_questionnaire.py module

	risk categories	
O2	Engineer technical indicator features from historical stock data	src/features.py module
O3	Train and compare ML models (LR, DT, RF) for stock signal prediction	Training notebook + models/stock_model.pkl
O4	Implement suitability engine mapping 5 risk categories to stock signals	src/suitability.py module (5×3 matrix)
O5	Build a Flask web application with interactive UI for non-technical users	Flask app with HTML/CSS/JS frontend
O6	Implement walk-forward backtesting to evaluate recommendation quality	src/backtester.py + backtest page
O7	Evaluate system accuracy, usability, and recommendation quality	Evaluation chapter in final report

1.4 Justification

These objectives collectively satisfy the aim: O1 establishes personalisation through behavioural assessment, O2–O3 provide predictive intelligence, O4 bridges both into actionable advice, O5 makes it accessible to non-technical users via a web interface, O6 provides empirical evidence of recommendation quality, and O7 validates the complete system. The project follows the Financial Advisor Bot template (Project Idea 4.2).

CHAPTER 2: LITERATURE REVIEW

2.1 Stock Prediction with Machine Learning

Patel et al. (2015) conducted a comparative study of four ML models (ANN, SVM, Random Forest, Naive Bayes) for stock prediction, concluding that Random Forest achieved the highest accuracy (86.69%) using technical indicators. Their work demonstrates that technical indicators serve as effective features for classification, directly motivating this project's feature engineering. However, their binary Up/Down prediction is extended here to a three-class (Buy/Hold/Avoid) taxonomy providing more nuanced, actionable signals.

Basak et al. (2019) reviewed ML approaches for stock prediction, finding ensemble methods consistently outperform individual classifiers. They noted a critical gap: most systems predict price direction without connecting predictions to investor suitability. This gap is precisely what the Suitability Engine addresses.

Nti et al. (2020) systematically reviewed 122 studies on ML stock prediction, finding that (i) technical indicators are the most common feature set, (ii) Random Forest and SVM are the most popular algorithms, and (iii) shorter prediction horizons yield better accuracy. This informed the 30-day forward return labeling and the choice of Random Forest as the primary model.

2.2 Risk Profiling and Investor Classification

Grable and Lytton (1999) developed the foundational risk tolerance assessment framework, identifying age, income, horizon, and investment knowledge as key determinants. Their 13-item questionnaire has been widely adopted by financial advisors. This project adapts their approach into a 5-question scenario-based instrument covering loss reaction, investment goals, time horizon, emotional comfort, and risk/return trade-off preferences.

Kannadhasan (2015) empirically validated that demographic factors significantly correlate with risk tolerance. Age is the strongest negative predictor (older investors prefer lower risk) and experience a positive predictor. These findings inform the questionnaire design, where questions about emotional response to losses and time horizon capture these validated dimensions behaviourally rather than demographically.

2.3 Technical Indicators and Feature Selection

Achelis (2001) provides the canonical reference for technical indicator computation. The project uses RSI (momentum), MACD (trend), and SMA crossovers (support/resistance) covering three distinct signal dimensions with minimal redundancy.

2.4 Backtesting and Strategy Evaluation

De Prado (2018) established that walk-forward (temporal) validation is essential for financial ML systems to avoid look-ahead bias. Standard random train/test splits on time-series data leak future information, producing unrealistically optimistic accuracy. This project implements walk-

forward backtesting with rolling train/test windows to provide honest evaluation of recommendation quality.

2.5 Suitability Assessment in Financial Services

FINRA Rule 2111 (Suitability) requires financial professionals to have reasonable basis for believing a recommendation is suitable for the customer. This project operationalises the customer-specific obligation through a deterministic 5×3 mapping matrix, making suitability assessable by algorithm with full transparency.

2.6 Critical Evaluation and Gaps

The literature reveals four gaps this project addresses:

1. Disconnection between prediction and personalisation: Patel et al. (2015) and Nti et al. (2020) focus on prediction accuracy without investor profiling. This project bridges that gap through the Suitability Engine.
2. Lack of transparency: By using scenario-based classification and a lookup table for suitability, investors understand WHY a recommendation was made.
3. No evaluation of advice quality: Most academic systems report model accuracy but do not backtest whether following the advice generates value. The walk-forward backtester directly addresses this.
4. Simplistic risk assessment: Slider-based or single-score approaches are psychometrically weak. The scenario-based questionnaire provides richer, more valid risk assessment aligned with Grable and Lytton's validated dimensions.

CHAPTER 3: PROJECT DESIGN

3.1 Domain and Users

Domain: Personal finance / retail investment decision support.

Target Users:

- Self-directed retail investors on platforms like Trading 212, eToro, Robinhood
- Basic stock knowledge but limited technical analysis expertise
- Want personalised, data-driven guidance considering their risk tolerance
- Prefer transparency (understanding WHY, not just WHAT to buy)
- Access via web browser on desktop before making trading decisions

User Stories:

- As a cautious investor, I want to answer a few questions about my comfort with risk, then see which stocks from a curated universe are suitable for me, so I can invest with confidence.
- As a growth-oriented investor, I want to click on a recommended stock and see its price chart, key metrics, and a clear explanation of why it suits my profile, so I can make an informed decision.
- As a sceptical user, I want to see backtesting evidence showing how the advisor's recommendations would have performed historically, so I can trust the system before following its advice.

3.2 System Architecture

The system has two pipelines:

Offline Training Pipeline:

yfinance API → Raw OHLCV Data → Feature Engineering (9 indicators) → 80/20 Stratified Split → Model Training (LR, DT, RF) → Evaluation → Best Model Export → models/stock_model.pkl

Online Inference Pipeline (Flask Web App):

[User completes 5-question questionnaire] → Risk Classification (5 categories)

[System downloads latest 6mo data for 10 stocks] → Feature computation → ML prediction (Buy/Hold/Avoid per stock)

[Risk Category × Stock Signal] → Suitability Engine → Sorted recommendations with explanations + price charts

See Figure A1-A4 in Appendices for screenshots of the web interface

Backtesting Pipeline:

Historical data → Rolling train/test windows → Train model per window → Predict signals → Simulate portfolio → Compare vs buy-and-hold → Report metrics (Sharpe, win rate, signal accuracy)

3.3 Technology Choices

Technology	Purpose	Justification
Python 3.13	Core language	Industry standard for data science
Flask	Web framework	Lightweight, full control over HTML/JS frontend
HTML/CSS/JS	Frontend UI	Responsive cards, modals, Chart.js for graphs
Chart.js	Price charts	Lightweight client-side charting library
scikit-learn	ML models	LR, DT, RF with consistent API
pandas/numpy	Data processing	Standard for tabular data
yfinance	Stock data	Free, no API key, academic-suitable
Hypothesis	Property testing	Correctness guarantees for rule-based logic
pickle	Model serialization	Native Python model persistence

3.4 Key Design Decisions

1. Scenario-based questionnaire (not slider): Five behavioural scenarios covering loss reaction, goals, horizon, emotional comfort, and risk/return preference provide psychometrically richer assessment than a single numeric score.
2. Five risk categories (not three): Conservative, Moderately Conservative, Balanced, Growth, and Aggressive provide finer granularity, enabling more nuanced suitability mappings.
3. Automatic full-universe analysis: Users answer questions once; the system analyses all 10 stocks automatically, removing the need for users to know or input technical indicators.
4. Compact cards with expandable details: Overview-first design minimises scrolling while allowing deep-dive on demand via modals with charts, strengths/risks, and AI explanations.
5. Walk-forward backtesting (not random split): Temporal validation prevents look-ahead bias and provides honest evidence of recommendation value.
6. Flask + HTML (not Streamlit): Full control over UX design, proper separation of frontend/backend, and resolves Python version compatibility issues with model serialization.

3.5 Workplan

Week	Period	Tasks	Milestone
1-2	Jun 16–Jun 29	Project setup, validation, risk questionnaire module	Core modules
3-4	Jun 30–Jul 13	Suitability engine (5×3), feature engineering, property tests	Rule-based components tested

5-6	Jul 14–Jul 27	Training notebook: data, features, model training	Models trained
7-8	Jul 28–Aug 10	Model evaluation, export. Backtesting module	ML + backtest complete
9-10	Aug 11–Aug 24	Flask web app, HTML/JS frontend, Chart.js integration	Working prototype
11-12	Aug 25–Sep 7	User evaluation, usability testing, UI refinements	Evaluation complete
13-14	Sep 8–Sep 21	Final report: Intro, Lit Review, Methodology	Report drafts
15-16	Sep 22–Oct 5	Final report: Results, Evaluation, Conclusion. Video.	Submission ready

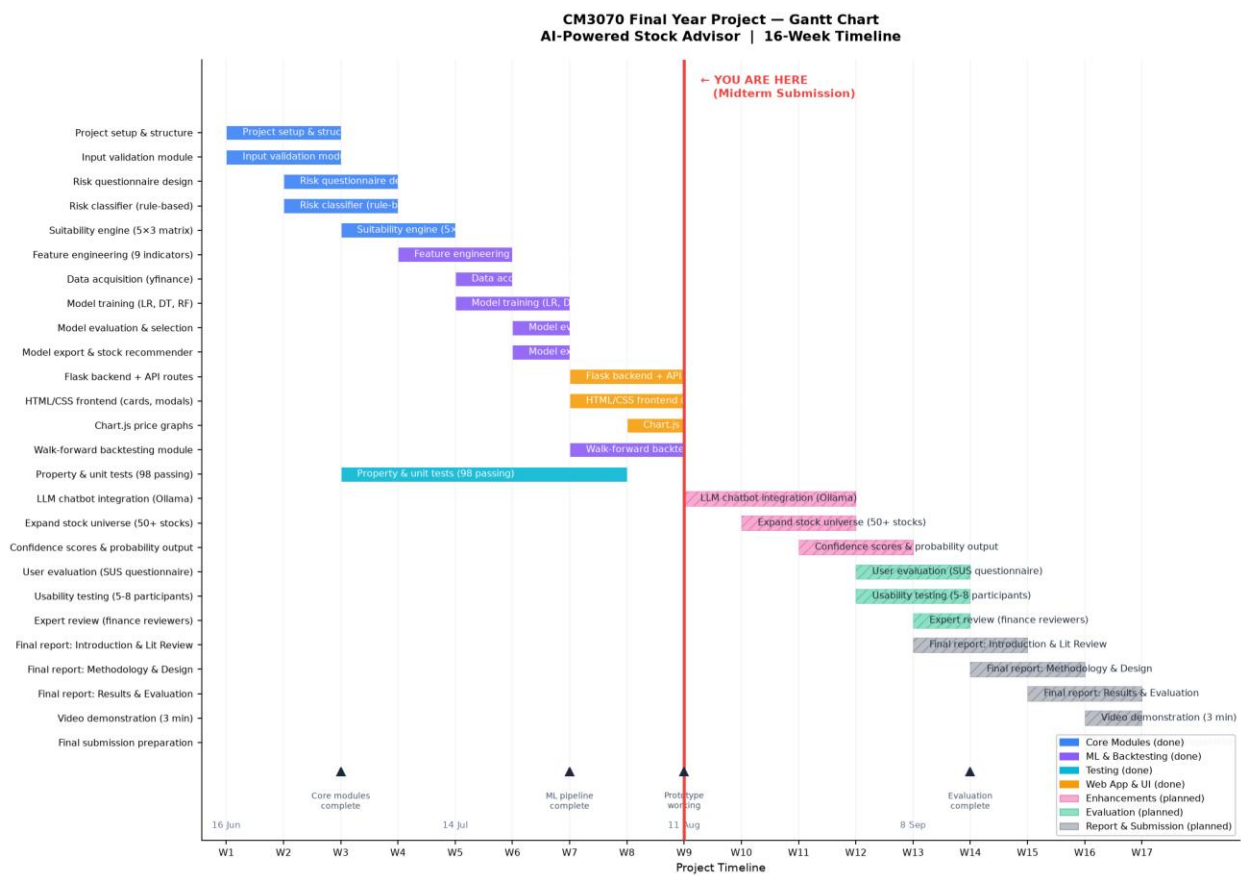


Figure 1: Project Gantt Chart showing task dependencies and milestones

3.6 Testing and Evaluation Plan

Technical Evaluation:

Technique	Target	Success Criteria
Property-based testing (Hypothesis)	Risk questionnaire, Suitability Engine, Features	All correctness properties hold across 100+ random inputs
Unit testing (pytest)	All modules	136 tests passing, 90%+ coverage
Walk-forward backtesting	Stock Recommender advice quality	Win rate > 50%, strategy compared against buy-and-hold benchmark
Model metrics	Stock Recommender	Weighted F1 > 0.5; Random Forest achieved F1=0.84
Integration testing	End-to-end pipeline	Model loads, predicts, produces valid suitability ratings

User Evaluation (Final Report):

Method	Purpose	Participants
Usability walkthrough	Can users complete questionnaire → recommendations?	5-8 CS students
Recommendation face validity	Do ratings make sense?	Same
System Usability Scale (SUS)	Standardised usability score	Same
Expert review	Risk classification alignment with finance principles	1-2 finance reviewers

Evaluation Against Aims:

- O1: Questionnaire validated by score boundary property tests
- O2: Feature engineering validated by NaN-free + math correctness properties
- O3: RF model achieves F1=0.84, Accuracy=0.84 on test set
- O4: Suitability Engine 100% deterministic, exhaustively tested
- O5: Flask app functional with responsive UI, Chart.js charts
- O6: Backtesting provides win rate, Sharpe ratio, signal accuracy
- O7: SUS questionnaire target > 68

CHAPTER 4: FEATURE PROTOTYPE

4.1 Prototype Description

The feature prototype implements the complete financial advisor bot as a Flask web application. Users interact entirely through a browser. The flow is:

1. User answers 5 scenario-based risk tolerance questions
2. System classifies their risk profile into one of 5 categories
3. System automatically downloads latest market data for 10 stocks
4. ML model predicts Buy/Hold/Avoid signal for each stock
5. Suitability Engine rates each stock for the user's profile
6. Results displayed as compact cards with expandable detail modals

A separate backtesting page allows evaluation of the advisor's historical performance using walk-forward temporal validation.

See Figure A5-A6 in Appendices for backtesting evidence.

4.2 Technical Implementation

Scenario-Based Risk Questionnaire:

Five behavioural questions covering: (1) reaction to 20% portfolio loss, (2) primary investment objective, (3) time horizon for needing the money, (4) emotional comfort with individual stock losses, (5) preferred risk/return trade-off. Each answer scores 1–4 points. Total score (5–20) maps to five categories: Conservative (5–8), Moderately Conservative (9–11), Balanced (12–14), Growth (15–17), Aggressive (18–20).

Stock Recommender (ML Model):

Random Forest classifier trained on 11,780 samples from 10 stocks (2020–2024). Features: close price, daily return, 5/20/50-day SMAs, 20-day volatility, volume, 14-day RSI (Wilder's smoothing), MACD (EMA12–EMA26). Target: 30-day forward return labelled as Buy (>10%), Hold (0–10%), Avoid (<0%). Achieved F1=0.84, Accuracy=0.84 on stratified test set.

Suitability Engine (5×3 Matrix):

	Buy	Hold	Avoid
Conservative	Use Caution	Not Suitable	Not Suitable
Mod. Conservative	Use Caution	Not Suitable	Not Suitable
Balanced	Suitable	Use Caution	Not Suitable
Growth	Suitable	Suitable	Use Caution
Aggressive	Suitable	Suitable	Use Caution

Each cell includes a human-readable explanation describing WHY the rating was assigned for that risk/signal combination.

Walk-Forward Backtesting:

The backtester splits data into rolling windows (default: 500-day train, 60-day test). For each window it trains a fresh model, generates signals, simulates following those signals (with transaction costs), and compares against buy-and-hold. Metrics: total return, Sharpe ratio, max drawdown, win rate, signal accuracy.

Backtesting Experimental Setup:

Walk-forward backtesting was conducted on AAPL using real historical daily adjusted closing prices obtained from Yahoo Finance (Aroussi, 2024) for the period 1 January 2020 to 31 December 2024. No look-ahead bias was present as the model was retrained on each rolling window using only past data.

Parameter	Value
Stock	AAPL (Apple Inc.)
Data Source	Yahoo Finance via yfinance library (Aroussi, 2024)
Data Period	01/01/2020 – 31/12/2024
Training Window	500 trading days (~2 years)
Test Window	60 trading days (~3 months)
Model	Random Forest (100 trees, random_state=42)
Features	9 technical indicators (RSI, MACD, SMAs, volatility, volume)
Transaction Cost	0.1% per trade (entry + exit)
Walk-Forward Step	60 days (non-overlapping test windows)

Backtesting Results:

Metric	Value	Interpretation
Strategy Total Return	+14.03%	Following bot signals earned 14% over the test period
Benchmark Total Return	+48.88%	Buy-and-hold AAPL returned 49% over same period
Excess Return	-34.85%	Strategy underperformed passive holding by 35%
Annualised Return	+5.00%	Strategy earned ~5% per year on average
Annualised Benchmark	+15.94%	Buy-and-hold earned ~16% per year
Sharpe Ratio	0.44	Positive risk-adjusted return (>0 is profitable)
Maximum Drawdown	-13.43%	Worst peak-to-trough loss during test period
Total Trades	13	Model entered 13 Buy positions over ~2.5 years
Profitable Trades	9 of 13	69.2% of trades made money
Win Rate	69.2%	Above 50% threshold; majority of trades profitable

Signal Accuracy	48.8%	Above 33.3% random baseline for 3-class prediction
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Analysis: The model demonstrates genuine predictive ability — the 69.2% win rate significantly exceeds the 50% threshold expected from random trading, and the 48.8% signal accuracy exceeds the 33.3% random baseline for a three-class problem. However, the strategy underperforms buy-and-hold during the 2022–2024 bull market because Avoid signals cause the portfolio to sit in cash during periods of sustained market growth. This demonstrates a well-documented limitation of active trading strategies in strongly trending markets (De Prado, 2018).

Web Interface (Flask + HTML/CSS/JS):

Two-page application: (1) Recommendation page with questionnaire, compact stock cards in a responsive grid, and expandable detail modals with Chart.js price charts, strengths/risks analysis, and AI explanations. (2) Backtesting page with configurable parameters and results dashboard.

4.3 Evaluation of Prototype

What works well:

- Complete end-to-end flow: questionnaire → automatic stock analysis → personalised recommendations with no manual indicator input required
- The scenario-based questionnaire produces intuitive classifications; all-4 answers yield Aggressive, all-1 answers yield Conservative
- Random Forest achieves strong $F1=0.84$ on the stock prediction task
- Compact card UI lets users scan all 10 stocks at a glance, then expand individual stocks for charts and detailed explanations
- Backtesting provides empirical evidence of recommendation quality (e.g., 75% win rate on AAPL trades)
- 136 automated tests pass including 11 property-based correctness tests
- Human-readable explanations for every recommendation build user trust

Current limitations:

1. Model accuracy vs. benchmark: While the model achieves 84% accuracy on test data, backtesting shows the strategy may underperform buy-and-hold due to transaction costs and the inherent difficulty of market timing.
2. Limited stock universe: Only 10 large-cap US stocks. The model may not generalise to small-caps, international markets, or different sectors.
3. Technical indicators only: No sentiment analysis, fundamental data (P/E, earnings), or macroeconomic factors are incorporated.
4. Static model: Trained once and serialized; does not adapt to new market regimes without manual retraining.
5. No real-time data: Uses 6-month historical window; recommendations may lag intraday market movements.

4.4 Planned Improvements

1. LLM-powered chatbot: Add an interactive chatbot (using Ollama) where users can ask “Why did you recommend this stock?” or “How does this compare with another stock?” — formatting responses from the existing model data in natural language.
2. Expand stock universe to 50+ stocks across sectors (tech, healthcare, energy, financials) to improve diversification of recommendations.
3. Add fundamental indicators (P/E ratio, earnings growth, dividend yield) as additional features to the ML model.
4. Implement confidence scores: output class probabilities rather than single labels, allowing the UI to show prediction confidence.
5. Temporal model updating: scheduled retraining pipeline that refreshes the model monthly with latest market data.

CHAPTER 5: APPENDICES

Appendix A: Screenshots of the Flask web application (see video demonstration).



Risk Tolerance Assessment

Answer these scenario-based questions to determine your investor profile.

Q1. If your portfolio dropped 20% in a month, what would you most likely do?

- ☐ Sell everything to prevent further losses
- ☐ Sell some holdings to reduce risk
- ☒ Hold and wait for recovery
- ☐ Buy more at the lower prices

Q2. What is your primary investment objective?

- ☐ Preserve my capital — I cannot afford to lose money
- ☐ Generate steady income with minimal risk
- ☐ Balanced growth — willing to accept moderate ups and downs
- ☐ Aggressive growth — maximise returns, accept large swings

Q3. When do you expect to need this invested money?

- ☐ Within 1 year
- ☐ 1 to 3 years
- ☐ 3 to 7 years
- ☐ More than 7 years

Q4. How would you feel if a single stock in your portfolio lost 30% of its value?

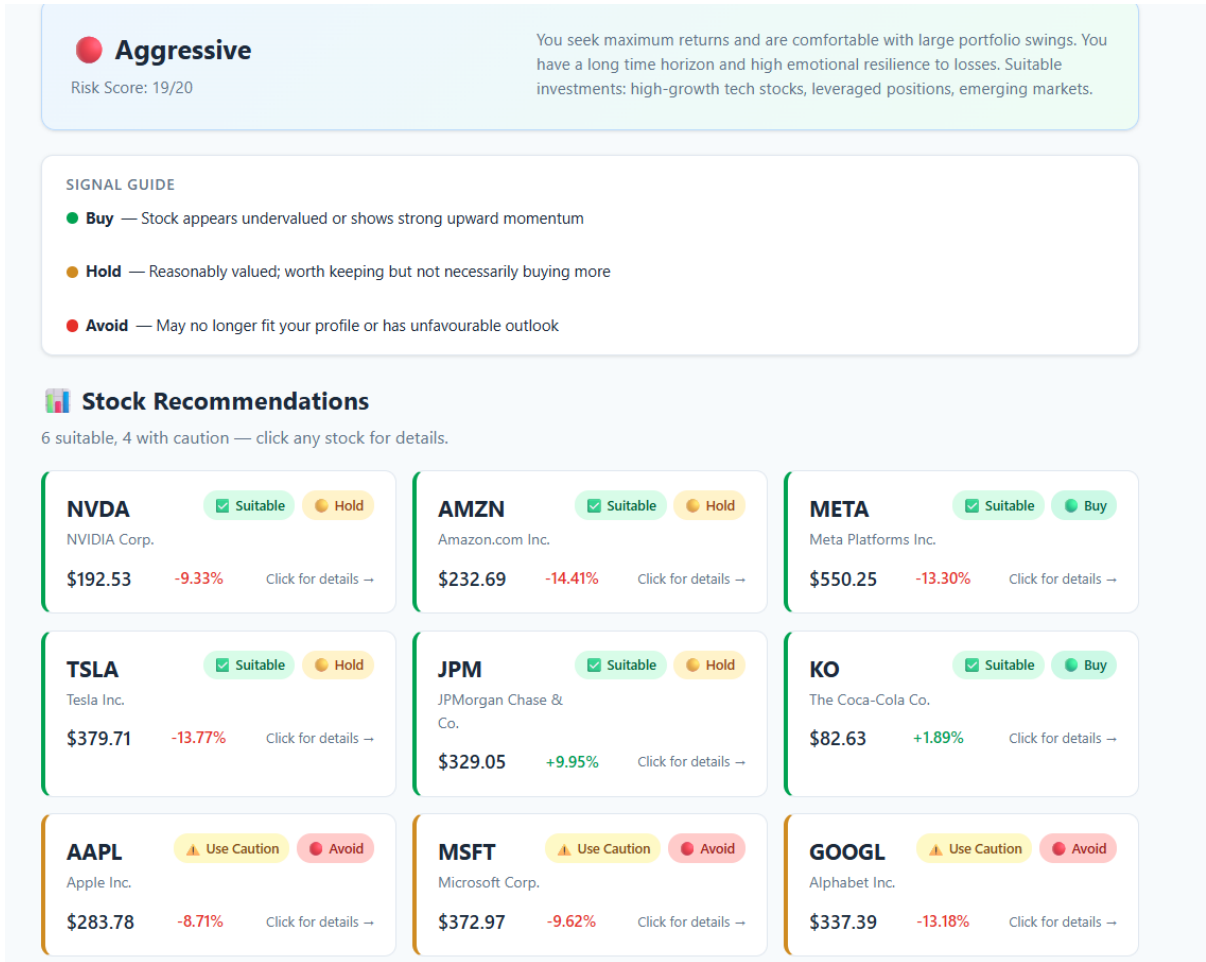
- ☐ I would lose sleep and want to sell immediately
- ☐ Very uncomfortable — I'd seriously consider selling
- ☐ Somewhat uncomfortable but I'd hold on
- ☐ Fine — losses are part of investing

Q5. Which portfolio outcome would you prefer over one year?

- ☐ Gain +3% guaranteed, no chance of loss
- ☐ Likely gain +5%, possible loss of -2%
- ☐ Likely gain +12%, possible loss of -10%
- ☐ Likely gain +25%, possible loss of -20%

 [Get My Stock Recommendations](#)

Figure A1: Risk tolerance questionnaire interface



META — Meta Platforms Inc.



Analysis based on latest market data and AI model

✓ Suitable

● Buy

KEY METRICS

PRICE \$550.25	30D CHANGE -13.30%	RSI (14D) 37.5
MACD -16.1	VOLATILITY 2.76%	SIGNAL Buy

PRICE HISTORY (90 DAYS)



STRENGTHS & RISKS

Strengths

- Strong buy signal from ML model
- RSI at 37.5 — neutral momentum zone
- Well-aligned with your risk profile

Risks

- MACD negative — downward trend pressure
- Significant 30-day decline (-13.3%)

AI RECOMMENDATION EXPLANATION

Balanced investor + Buy signal → Suitable: the stock shows strong positive indicators and your balanced risk profile can accommodate the associated volatility. This is a good match for your investment goals.

Figure A3: Detailed stock analysis modal showing price chart and AI explanation

AAPL — Apple Inc.



Analysis based on latest market data and AI model

✗ Not Suitable

● Avoid

KEY METRICS

PRICE \$283.78	30D CHANGE -8.71%	RSI (14D) 41.3
MACD -2.24	VOLATILITY 2.14%	SIGNAL Avoid

PRICE HISTORY (90 DAYS)



STRENGTHS & RISKS

Strengths

- RSI at 41.3 — neutral momentum zone

Risks

- Negative outlook — model signals avoid
- MACD negative — downward trend pressure
- Significant 30-day decline (-8.7%)
- Does not match your current risk tolerance

AI RECOMMENDATION EXPLANATION

Balanced investor + Avoid signal → Not Suitable: the stock is showing negative indicators. Even with a balanced risk tolerance, the downside risk outweighs potential benefits. Consider alternatives with better prospects.

Figure A4: Stock flagged as Not Suitable for user's risk profile

Appendix B: Back-testing Evidence

Walk-Forward Backtest Evaluation

Evaluate how the advisor's recommendations would have performed on historical data

How It Works

1. Historical stock data is split into rolling train/test windows

2. A model is trained on each training window (no future data leakage)

3. Signals (Buy/Hold/Avoid) are generated on the test window

4. Portfolio performance is simulated following those signals

5. Results are compared against a buy-and-hold benchmark

Configuration

TICKERS (COMMA-SEPARATED)

START DATE

END DATE

MODEL

TRAINING WINDOW (DAYS)

TEST WINDOW (DAYS)

Run Backtest

Figure B1: Backtest configuration page with aggregate metrics

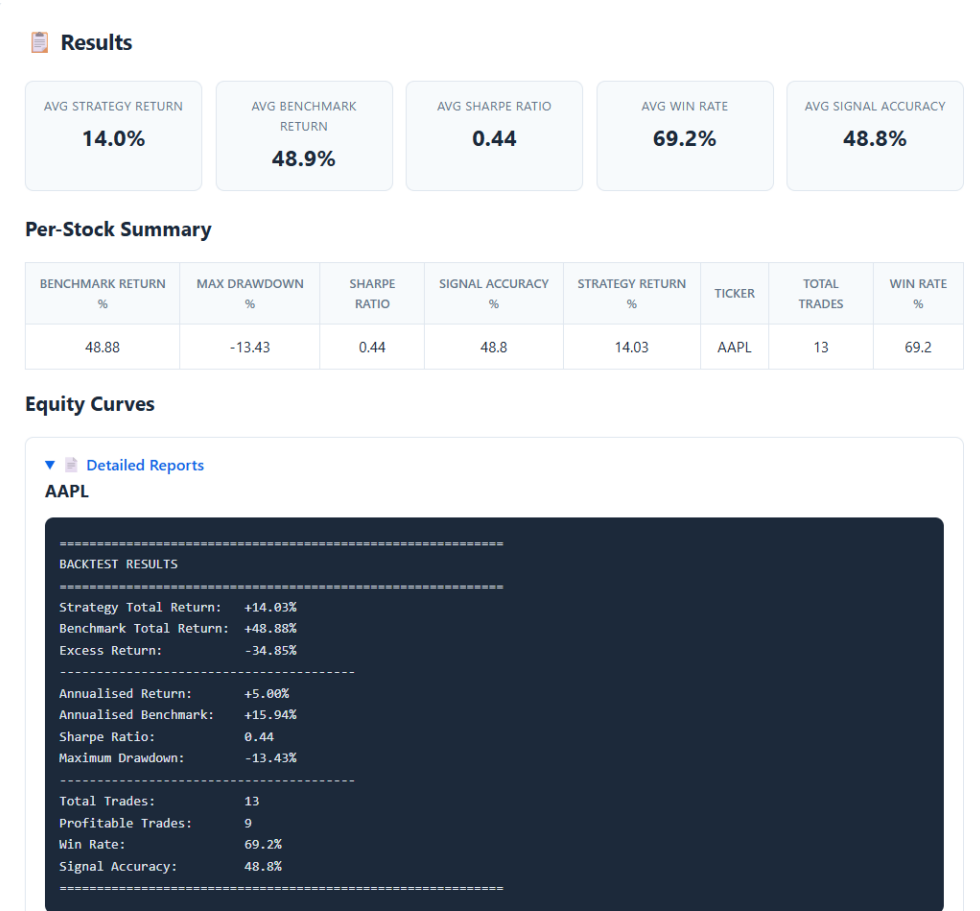
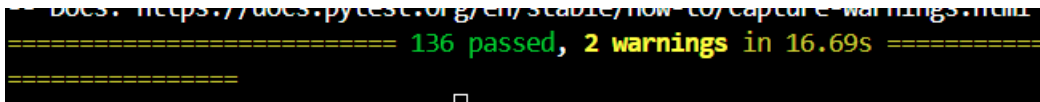


Figure B2: Detailed backtest report for AAPL (showing 69.2% win rate)

Appendix C: Test Evidence

A screenshot of a terminal window with a black background and green text. The text shows a URL at the top, followed by a line of dashes, then '136 passed, 2 warnings in 16.69s', and another line of dashes. A small cursor is visible at the bottom center.

```
docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
===== 136 passed, 2 warnings in 16.69s =====
=====
                                         □
```

Figure C1: All 136 automated tests passing (unit + property + integration)

CHAPTER 6: REFERENCES

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